



FIRST SEMESTER, 2025–2026
Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : MEF 212
Course Title : Fluid Mechanics
Instructor-in-Charge : Prof. Shyam Sunder Yadav
Instructors : Prof. Shyam Sunder
Yadav, Prof. Gaurav Sharma, Nancy, Rahul, Rahul
Bharat, Christo

1. Course Description:

This course covers the following topics: Mathematical preliminaries; Fluid statics and forces on submerged bodies; Integral relations for a control volume; Mass, Momentum and Energy Conservation; Reynolds Transport Theorem; Differential relations for fluid flow, Analytical solution of Navier-Stokes equation; Dimensional analysis and similarity; Inviscid potential flows; Bernoulli's Equation; Internal and external viscous flows; Basic compressible flows.

2. Scope & Objectives of the Course:

The objectives of the course are to learn about the following topics:

- Introduction to fluid mechanics and establish its relevance in mechanical engineering.
- Derivation of the equations governing mass, momentum and energy transport.
- Application of the governing equation to solve simple fluid mechanics problems.
 - Non-dimensionalization of governing equations and identification of dimensionless numbers, Modeling and similarity concepts.
- Two dimensional, inviscid, potential flows; Stream function, Velocity potential.
 - Analyze laminar and turbulent flows inside pipes, External flow past immersed bodies, Intro- duction to Boundary layer theory
 - One dimensional, compressible flow inside ducts of variable cross-section, normal and oblique shocks.

3. Textbooks:

TB1. Fluid Mechanics, Frank M. White, 8th Edition, Tata McGraw Hill India

References:

Ref1. Introduction to Fluid Mechanics, Robert W. Fox & Alan T. McDonald, 10th Edition, John Wiley Publications India.

Ref2. Mechanics of fluids, Potter, Wiggert and Ramadan, 5th Edition, Cengage India.

4. Course Plan:

The following is the lecture plan:

Week no.	Module No.	Lecture No.	Reference	Learning Outcomes
1.	1. Mathematical preliminaries	1-3	Class notes	Scalars, Vectors, Tensors, Dot and Cross products, Gradient, Divergence and Curl, Forces on a surface, Gauss' and Stokes' theorems
2, 3.	2. Fluid Statics	4-7	TB1 2.1-2.7	Introduction, Pressure on submerged surfaces, Pressure distribution under rigid body motion
3.	3. Flow theorems	8-9	Class notes	Lagrangian and Eulerian representation, Reynolds Transport Theorem, Material derivative
4, 5.	4. Integral Relations for a Control Volume	10-15	TB1 3.1-3.6	Conservation of mass, momentum, angular momentum and energy for a control volume, Applications to simple flow situations
6, 7.	5. Differential relations for fluid flow	16-20	TB 1 4.1-4.6	Conservation of mass, momentum and energy for a differential fluid element, Newtonian and Non-Newtonian fluids, Navier-Stokes equations and their exact solutions, Boundary conditions
7, 8.	6. Dimensional Analysis and Similitude	21-24	TB 1 5.1-5.5	Buckingham Pi theorem, Non-dimensional groups for fluid flow, Non-dimensionalization of Navier-Stokes equations
9, 10.	7. Incompressible Inviscid flows	25-28	Class notes	Euler equation, Bernoulli equation, Stream function, Velocity potential, Elementary inviscid flows: uniform, source, sink, vortex, Superposition

11, 12.	8. Viscous Flow in Ducts	29-33	TB 1 6.1-6.7	Viscous, laminar and turbulent flows in pipes
12, 13.	9. Flow past immersed bodies	34-38	TB 1 7.1-7.5	Flow past immersed bodies, Introduction to Boundary layer theory, Drag and Lift, Boundary layer separation
13, 14.	10. Compressible flow	39-42	TB 1 9.1-9.6	Mach number, Speed of sound, Steady isentropic, adiabatic flow in variable area ducts, Normal shock, Oblique shock

5. Evaluation Scheme:

Component	Weightage (%)	Duration	Date & Time	Comments
Tutorial tests	20	40 mins.	Tutorial class	Closed book
Objective Quizzes	10	30 mins.	Tutorial class	Closed Book
Mid-Sem. exam.	25	90 mins.	As per timetable	Open Book
Compre. exam.	45	180 mins.	As per timetable	Open Book

NC Criterion: As per AUGSD instructions

6. Tutorials:

There will be five evaluative tutorial tests, best four will be considered. Syllabus for each will be based on individual chapters covered in the lectures. There will be no makeup for these evaluative tutorial tests.

7. Objective quizzes:

There will be two objective quizzes. One will be conducted just before the mid sem. exam. (based on the mid sem. syllabus) and the other will be conducted just before compre. exam. (based on the post mid sem. syllabus) . Best one will be considered. No makeup allowed for these objective quizzes.

8. Make up policy:

Mid sem. and comprehensive exam. make-ups will be granted only to genuine cases. For cases related to illness, prior information and proper documentary evidence are essential. Diarrhea,

headache etc. cases just before the exams. will not be considered. No make-up for tutorial tests and objective quiz components.

9. Office Consultation Hour:

Will be announced in the class. Also, feel free to email me to seek a prior appointment.

10. Notices:

Notices concerning the course will be sent through emails only. The students should check their BITS email regularly.

11. Open Book Exams.:

Photocopies of other's hand written notes and printouts of solution manuals will not be allowed in the exam.

Dr. Shyam Sunder Yadav
Instructor-in-charge
MEF 212, Fluid Mechanics